

Lab 10: Implement Dijkstra's Shortest Path Algorithm

Theory:

Dijkstra's algorithm finds the shortest path from a starting node to all other nodes in a graph with non-negative edge weights. It uses a priority queue to greedily select the closest node that has not yet been processed, performing a relaxation step on its outgoing edges.

Space Complexity: $O(V)$

Expressing Complexity of cases

Data Structure	Time Complexity	Condition
Array	$O(V^2)$	Dense graph
Min-priority Q	$O(E + V \log V)$	Sparse graph

Algorithm

1. Mark all nodes unvisited. Create a set of all the unvisited nodes.
2. Assign to every node a tentative distance value: set it to zero for an initial node and to infinity for all other nodes.
3. For the current node, consider all of its unvisited neighbors and calculate their tentative distances through the current node.
4. When done considering all the unvisited neighbors of the current node, mark the current node as visited.

Pseudocode

```
BEGIN
  FOR each vertex v in Graph DO
    dist[v] ← ∞
    prev[v] ← undefined
    add v to Q
  END FOR
  dist[source] ← 0
  WHILE Q is not empty DO
    u ← vertex in Q with min dist[u]
    remove u from Q
    FOR each neighbor v of u IN Q DO
      alt ← dist[u] + length(u, v)
      IF alt < dist[v]
        THEN dist[v] ← alt
        prev[v] ← u
      END IF
    END FOR
  END WHILE
END
```

Conclusion:

The implementation successfully validates Dijkstra's algorithm for finding the single-source shortest paths in a weighted graph. Utilizing a greedy approach with edge relaxation, the algorithm accurately distinguished the minimum distance from the Source node 'A' to all other accessible nodes.

Given the constructed graph array, the shortest distances mapped out exactly as expected: $A \rightarrow A$ (0), $A \rightarrow B$ (2), $A \rightarrow C$ (5), $A \rightarrow D$ (6), and $A \rightarrow E$ (7). The algorithmic execution correctly bypassed longer overlapping edge paths (such as $A \rightarrow B \rightarrow D$ representing a cost of 10) in favor of the direct path $A \rightarrow D$ at a cost of 6. This confirms the efficacy of Dijkstra's protocol in achieving an optimal uniform shortest path in unconstrained positive-weighted networks.